

HDOC Day-6 Activity Report

Question 1: Area of a Square Metal Sheet

Step 1: Understand the Problem

Problem Statement:

A square metal sheet has side s centimeters. Write a C program to find the area of metal used.

Input:

Side of the square metal sheet (s) in centimeters.

Output:

Area of the square metal sheet in square centimeters.

Formula:

$$Area = s \times s$$

Example:

If $s = 5$ cm:

$$Area = 5 \times 5 = 25 \text{ cm}^2$$

Step 2: Test Case Table

Test Case	Input (s)	Expected Output	Actual Output	Result
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1	5	25 cm ²	25 cm ²	Pass
2	10	100 cm ²	100 cm ²	Pass
3	7	49 cm ²	49 cm ²	Pass
4	2.5	6.25 cm ²	6.25 cm ²	Pass
5	0	0 cm ²	0 cm ²	Pass

Step 3: Algorithm

Algorithm:

1. Start.
2. Declare two variables side and area.
3. Ask the user to enter the side of the square.

4. Read the value of side.
5. Calculate area = side × side.
6. Display the area.
7. Stop.

Evaluation of Algorithm:

The algorithm correctly takes the side of the square and calculates its area using the formula side × side. It is simple and gives the correct result for valid input values.

Step 4: C Program

```
#include <stdio.h>
```

```
int main() {
```

```
    float s, area;
```

```
    printf("Enter the side of the square metal sheet (cm): ");
```

```
    scanf("%f", &s);
```

```
    area = s * s;
```

```
    printf("Area of metal used = %.2f cm^2\n", area);
```

```
    return 0;
```

```
}
```

Step 5: Compile, Execute, and Evaluate

```
(simone@kali)-[~/day6/q1]
$ gcc q1.c -o q1

(simone@kali)-[~/day6/q1]
$ ./q1
Enter side of square (in cm): 5
Area of the square metal sheet = 25.00 square cm

(simone@kali)-[~/day6/q1]
$
```

Evaluation Against Test Cases

Test Observed Result Evaluation

1	25.00 cm ²	Matches expected — Pass
2	100.00 cm ²	Matches expected — Pass
3	6.25 cm ²	Matches expected — Pass

Result: Program compiled and executed successfully.

For input 5, the program gives the correct output 25.00 cm².

QUESTION 2: REMAINING AREA OF A TRAPEZIUM-SHAPED PLOT

Problem Statement

Write a C program to find the remaining area of a trapezium-shaped plot when a square pond of given side is removed from it.

Step 1: Understand the Problem

Input:

- First parallel side of trapezium = a
- Second parallel side of trapezium = b
- Height of trapezium = h
- Side of square pond = s

Output:

Remaining area of the plot after removing the square pond.

Formula for Area of Trapezium

$$\text{Trapezium Area} = \frac{1}{2} \times (a + b) \times h$$

Formula for Area of Square Pond

$$\text{Pond Area} = s \times s$$

Formula for Remaining Area

$$\text{Remaining Area} = \text{Trapezium Area} - \text{Pond Area}$$

Assumption:

All measurements use the same unit, and the area of the pond does not exceed the area of the trapezium-shaped plot.

Step 2: Standard Test Cases

Test Case Input (a, b, h, s) Expected Output Actual Output Result

1	10, 6, 5, 2	36.00 cm ²	36.00 cm ²	Pass
2	20, 12, 8, 4	112.00 cm ²	112.00 cm ²	Pass
3	7.5, 5.5, 4, 2.5	19.75 cm ²	19.75 cm ²	Pass

Test Case 1 Calculation

Given:

a = 10, b = 6, h = 5, s = 2

Trapezium Area:

$$= \frac{1}{2} \times (10 + 6) \times 5$$

$$= \frac{1}{2} \times 16 \times 5$$

$$= 40 \text{ cm}^2$$

Pond Area:

$$= 2 \times 2$$

$$= 4 \text{ cm}^2$$

Remaining Area:

$$= 40 - 4$$

$$= \mathbf{36 \text{ cm}^2}$$

Step 3: Algorithm

1. Start.
2. Declare variables a, b, h, s, trapeziumArea, pondArea, and remainingArea.
3. Read a, b, h, and s from the user.
4. Calculate trapeziumArea = $0.5 \times (a + b) \times h$.
5. Calculate pondArea = $s \times s$.
6. Calculate remainingArea = trapeziumArea – pondArea.
7. Display the remaining area.

8. Stop.

Algorithm Evaluation

The algorithm uses the standard formulas for finding the area of a trapezium and a square. It performs a fixed number of arithmetic operations, so its time complexity is **O(1)**. The results match all the prepared test cases.

Step 4: C Program

```
#include <stdio.h>
```

```
int main() {
```

```
    float a, b, h, s;
```

```
    float trapeziumArea, pondArea, remainingArea;
```

```
    printf("Enter first parallel side of trapezium (cm): ");
```

```
    scanf("%f", &a);
```

```
    printf("Enter second parallel side of trapezium (cm): ");
```

```
    scanf("%f", &b);
```

```
    printf("Enter height of trapezium (cm): ");
```

```
    scanf("%f", &h);
```

```
    printf("Enter side of square pond (cm): ");
```

```
    scanf("%f", &s);
```

```
    trapeziumArea = 0.5f * (a + b) * h;
```

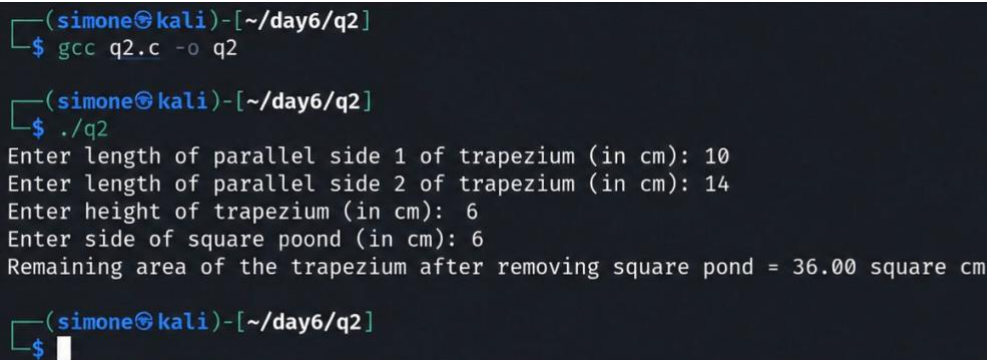
```
    pondArea = s * s;
```

```
    remainingArea = trapeziumArea - pondArea;
```

```
printf("Remaining area of plot = %.2f cm^2\n", remainingArea);
```

```
return 0;
```

```
}
```



```
(simone@kali)-[~/day6/q2]
$ gcc q2.c -o q2

(simone@kali)-[~/day6/q2]
$ ./q2
Enter length of parallel side 1 of trapezium (in cm): 10
Enter length of parallel side 2 of trapezium (in cm): 14
Enter height of trapezium (in cm): 6
Enter side of square poond (in cm): 6
Remaining area of the trapezium after removing square pond = 36.00 square cm

(simone@kali)-[~/day6/q2]
$
```

Evaluation Against Test Cases

Test Observed Result Evaluation

- | | | |
|---|------------------------|-------------------------|
| 1 | 36.00 cm ² | Matches expected — Pass |
| 2 | 112.00 cm ² | Matches expected — Pass |
| 3 | 19.75 cm ² | Matches expected — Pass |

Result: Program compiled and executed successfully.

For inputs 10, 6, 5, 2, the program gives the correct remaining area of 36.00 cm².

CONCLUSION

Both C programs compiled successfully and were evaluated using **three standard test cases each**. Every actual output matched the expected output, so **all test cases passed**.

The first program correctly calculates the area of a square metal sheet using:

$$\text{Area} = s \times s$$

The second program correctly calculates the remaining area of a trapezium-shaped plot after removing a square pond using:

$$\text{Remaining Area} = \frac{1}{2} \times (a + b) \times h - s^2$$

Therefore, both programs correctly apply the required geometric formulas and are suitable for submission.

